

Review article

A comprehensive overview of the alignment between platoon control approaches and clustering strategies

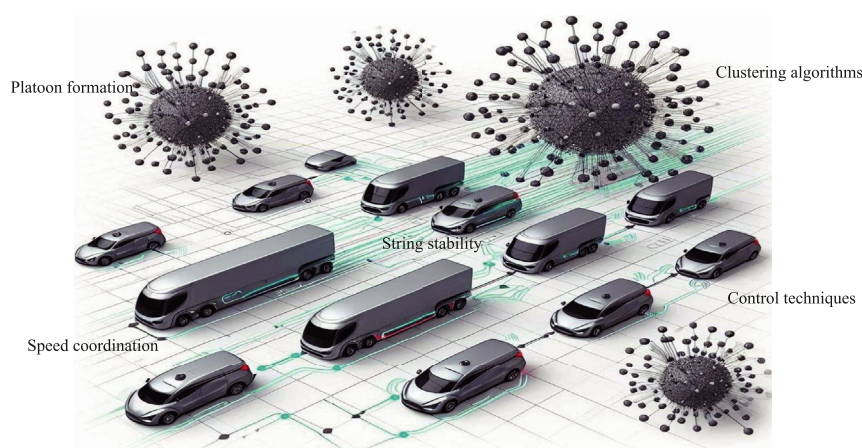
M. Nandhini^{*}, M. Mohamed Rabik^{**}

Department of Mechatronics Engineering, College of Engineering and Technology, SRM Institute of Science and Technology, SRM Nagar, Kattankulathur, 603203, Kanchipuram, Chennai, TN, India

HIGHLIGHTS

- Platooning emphasizes maintaining a desired formation through automatic speed adjustments within a group of vehicles.
- Clustering algorithms can optimize vehicle platooning systems.
- Effective control techniques ensure seamless communication, synchronization, and string stability.
- These techniques advance wireless-connected autonomous vehicles.

GRAPHICAL ABSTRACT



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ABSTRACT

Platoon formation focuses on effectively coordinating the speeds of vehicles within a group, with automatic speed adjustments for each vehicle to maintain a desired formation. The implementation of appropriate control techniques in platooning is crucial to achieve efficient vehicle coordination to facilitate seamless communication and synchronization among vehicles. The platoon functions as a cluster, where vehicles within the platoon are treated as nodes. This study presents an idea for implementing clustering strategies in a platoon with a focus on achieving string stability by decreasing disturbances and variations in vehicle speed and position. This also involves an in-depth analysis of clustering algorithms to identify the most suitable approach for integration into vehicle platooning, specifically for network analysis purposes. The investigation of various control techniques and clustering algorithms aims to optimize the performance and functionality of platooning systems contributing to the advancement of wireless-connected autonomous vehicles and their transformative potential in transportation.

^{*} Corresponding author.

^{**} Corresponding author.

E-mail addresses: nandhinm@srmist.edu.in (M. Nandhini), mohamedm3@srmist.edu.in (M.M. Rabik).

1. Introduction

Platooning, which involves driving a group of vehicles closely together, offers numerous benefits such as increased road capacity, improved traffic flow, enhanced highway safety, decreased energy consumption, and decreased pollution [1]. In a platoon, vehicles communicate with each other, with a lead vehicle controlling speed and direction while the following vehicles respond accordingly. Inter-vehicle communication and onboard sensors enable coordinated movement within the platoon. Platooning is an essential feature of self-driving cars, enabling faster, safer, and more efficient travel. Transportation planners optimize merge operations to form platoons effectively contemplating factors such as traffic conditions, vehicle speed adjustment strategies, and mergeable vehicle density. Trajectory planning techniques play a crucial role in controlling lateral motion in platooning and autonomous driving. Research in this area categorizes approaches based on platooning or lane changing, traffic scenarios, and cost functions [2]. Insight the convergence of platoons and their impact on traffic congestion is vital for effective management, as merging multiple platoons into a single entity can lead to jams [3]. Platooning heavy-duty vehicles on highways enhances energy and transport efficiency by increasing capacity and reducing air drag [4]. Cooperative driving systems rely on algorithms and wireless communication technologies to control vehicle motion based on the behaviour of neighbouring vehicles. Connectivity and autonomy in automated transportation offer potential benefits. Vehicle platooning entails a cluster of interconnected automated vehicles (CAVs) moving in unison at a synchronized speed, while adhering to a specific distance from the lead vehicle. Control strategies focus on overseeing transitional platoon maneuvers, including merge/split and lane changes, while considering relevant vehicle dynamics [5]. In the assessment of vehicle platoon performance and effectiveness, particular attention is given to the information flow topology and communication quality, considering various communication impairments such as restricted range, random packet losses and varying time delays. The research commences with an analysis of a scenario in which each vehicle employs information on position, velocity, and acceleration from both preceding and following vehicles. It establishes suitable environments to ensure the internal stability of the vehicle platoon system, employing matrix polynomial stability and matrix eigenvalue perturbation theory.

Furthermore, the research explores the influence of random packet losses on platoon control. It states that the effectiveness of platoon management is contingent on the occurrence of successful packet reception or the duration until a new packet is received. A maximum threshold for communication delays across all links is determined, allowing for the preservation of internal stability within the vehicle platoon, even when the delays vary within the specified limit. The analysis also includes an evaluation of L2-string stability using transfer functions across various communication scenarios, such as ideal communication, uniform delays, and constant delays. Effective communication of control information among vehicles in platooning significantly enhances both traffic safety and efficiency [6]. Proximity-based device-to-device (D2D) communications with elevated transmission rates are seen as potential solutions for facilitating the efficient and swift sharing of control information. To achieve this objective, a two-stage platoon formation algorithm is proposed. It involves evaluating potential platoon leaders to establish stable platoons. This approach reduces spectrum resource overhead and improves platoon safety. Moreover, a communication mode of predecessor-following is deployed, utilizing D2D technology for the high-speed transmission of control information among vehicles. To improve efficiency in resource allocation, two mechanisms are introduced. The first mechanism, named time division-based intra-platoon allocation, efficiently allocates resources for D2D users within the platoon. The second mechanism, minimum rate guaranteed inter-platoon allocation, optimizes rates for cellular users while ensuring effective resource allocation. Together, these mechanisms and the platoon formation algorithm enhance spectrum resource

utilization and ensure platoon stability. In summary, the proposed approach provides substantial advantages in terms of both resource utilization and the preservation of a secure and stable platoon [7]. A group of vehicles going at a steady pace utilizing modern communication and control technologies to maintain a safe distance is called vehicle platooning. This method boosts fuel economy, road safety, and congestion reduction. This configuration enhances traffic efficiency by establishing virtual connections among the vehicles. However, maintaining reliable communication within the platoon, particularly when integrating heavy duty or specialized vehicles presents challenges. A Spectrum Sensing Scheduling (SSS) scheme is suggested to manage communication resources in platooning and tackle associated challenges. The scheme incorporates spectrum sensing, platoon control error, and platoon communication delay models to evaluate delays in the platoon under SSS. The algorithm aims to minimize platoon delay, thereby improving overall platoon performance. The proposed SSS scheme exhibits significant reductions in platoon communication delay and enhances platoon safety compared to existing schemes. By effectively managing communication resources, the scheme improves the reliability and efficiency of communication within the platoon [8].

Through the integration of an on-board unit (OBU) for wireless communication, vehicles can form a vehicular ad hoc network (VANET), thereby improving road safety, passenger comfort, and communication with roadside infrastructure. VANETs enable vehicles to collect information from neighbouring vehicles and roadside equipment, facilitating improved communication and coordination. However, routing messages in VANETs can be challenging due to high mobility and dynamic network conditions. To address these challenges, clustering techniques can be employed in VANETs. These techniques involve grouping vehicles into virtual clusters based on various characteristics such as density, velocity, and location. Clustering ensures cluster stability and helps maintain the overall network structure, enabling efficient and reliable message routing within the VANET.

Vehicle Ad-Hoc Networks need communication. Traffic conditions, road challenges, and future events may be sent between cars using V2V communication. This real-time data interchange promotes cooperative driving and road safety. V2I communication lets cars communicate with infrastructure like traffic signals, road signs, and toll booths, improving traffic flow and giving drivers useful information. V2I communication facilitates communication between vehicles and roadside infrastructure, providing internet access, and supporting additional services and functionalities. By integrating these communication capabilities, VANETs contribute to improved road safety, enhanced passenger comfort, and efficient communication between vehicles and the surrounding infrastructure.

In Fig. 1, the schematic of homogeneous vehicle platooning is depicted, illustrating vehicles S_0 to S_N , distances D_1 to D_N between vehicles, and vehicle lengths L_1 to L_N . Advancements in modern communication technologies like radar-sensing systems, Bluetooth, GPS, wireless communication, drive-by-wire steering and throttle enable computers to assume control over vehicles. The emergence of 5G communication holds promise in handling the large volume of data required for safe platooning. However, wireless communication in platooning poses challenges such as information delay and diverse interaction topologies. This paper investigates various control approaches and equivalent network techniques to enhance performance efficiency in platooning.

2. Controls for platooning

2.1. Cruise control methods

2.1.1. Connected cruise control

The research concentrated on analysing the dynamics of connected vehicle systems (CVSs), encompassing both connected cruise control (CCC) and traditional vehicles. The objective was to investigate how

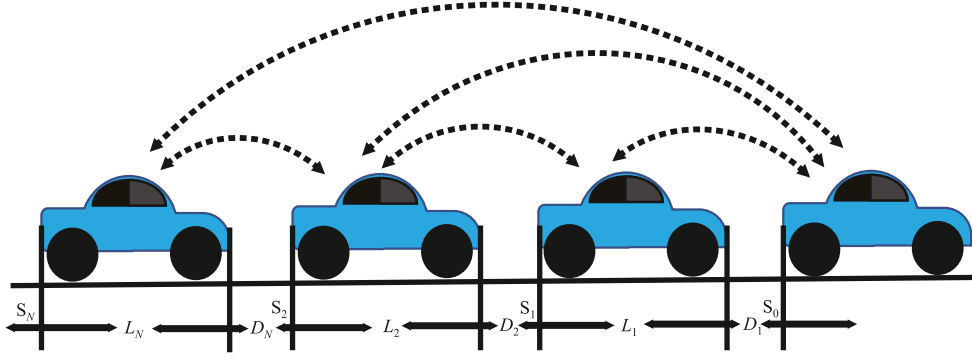


Fig. 1. Schematic of vehicle platooning.

various connectivity configurations and information delays affect the system. A comprehensive framework was introduced to facilitate CCC design, considering information delays and accommodating various connectivity structures. Stability criteria, dependent on delay considerations, were established to ensure stability in the overall system and the head-to-tail arrangement of vehicles in CVSSs. An approach based on motifs was suggested to facilitate the development of large-scale networks with modular and scalable designs. The results highlighted the advantages of Connected Cruise Control (CCC) vehicles in enhancing traffic efficiency. Nevertheless, it was noted that merely strengthening the quantity of CCC vehicles may not enhance system robustness unless the design of the control gains is appropriately addressed [9–12].

2.1.2. Adaptive cruise control

Adaptive Cruise Control (ACC) stands as a well-established method designed to improve the driving experience. It entails outfitting vehicles with a system that enables them to autonomously track the vehicle in front by utilizing relative state information from onboard sensors. Radar sensors placed on the front bumper of each vehicle measure the distance between vehicles, and wireless communication enables interaction between each automobile in the platoon and its leading vehicle. Ensuring practical string stability is essential for the safe and efficient operation of platoons incorporating vehicles equipped with ACC systems. It ensures coordinated behaviour and stability by maintaining safe following distances through sensor inputs and vehicle communication. This concept aims to reduce traffic congestion, enhance fuel efficiency, and address challenges like varying traffic densities, communication delays, and system uncertainties. Practical string stability combines adaptive control algorithms, communication protocols, and platoon management strategies to optimize performance and ensure safe coordination between vehicles in real-world scenarios [13].

2.1.3. Cooperative Adaptive Cruise Control

Cooperative Adaptive Cruise Control (CACC)-enabled platooning is an emerging technology that has attracted considerable attention. It allows vehicles to closely follow each other, offering benefits such as increased street capacity, reduced fuel consumption and emissions, and improved driver comfort and safety. Future research focuses on addressing challenges related to multi-objectiveness and individualization in platooning coordination. The proposal suggests implementing CACC systems to increase platoon performance, allowing automobiles to exchange real-time state information to achieve closer inter-vehicle distances. The Adaptive Optimal Control approach addresses uncertainties and heterogeneity in CACC systems, improving stability and performance. Heterogeneous platooning and communication losses in Vehicular Ad-Hoc Networks may be addressed using adaptive switched control. This method dynamically switches control modes depending on platoon vehicle communication reliability. Platooning vehicles, equipped with CACC, offer advantages in terms of comfortable driving, safety, and transportation efficiency. Both distributed and centralized control

frameworks are investigated for platooning control, considering platoon uncertainty. The string stability of CACC platoons is influenced by complex driving situations and unusual driving behaviours. A framework is implemented that integrates driving state recognition, platoon operations, risk estimate to reduce disturbances and enhance safety. Criteria are put forth to attain string stability in mixed traffic platoons, guaranteeing the boundedness of motion states and overall safety of the platoon. A human centered CACC system improves driver comfort while maintaining mixed traffic string stability. To fully benefit from CACC, optimizing communication and minimizing unnecessary transmissions is important to establish a stable platoon [14–16]. Fig. 2 represents the communication structure used for cruise control among the vehicles.

2.1.4. Leader and predecessor following control

The coordinated control of a vehicle platoon requires an exploration of various communication structures and their influence on platoon stability. This investigation delves into various specific communication structures. Employing an adaptive control law, the study estimates parameter uncertainties and utilizes a decentralized approach to ensure the platoon stability [17]. Within the context of platooning, the control architecture employed is Leader and Predecessor Following (LPF). Nevertheless, conventional LPF strategies depend on detailed knowledge of the spacing policies of predecessor vehicles which may not be practical in ad hoc platoons. To address this limitation, a virtual spacing policy is introduced as a substitute for the unknown spacing policies. When this virtual spacing policy is integrated into a two-degree-of-freedom LPF controller, it makes it easier to create ad hoc platoons with vehicles that have different spacing rules [18]. The control architecture of autonomous vehicles in multi-brand ad hoc platoons also embrace the LPF approach. In such instances, having precise knowledge of the spacing policies of leader vehicles may not be attainable. To tackle this challenge, a 2 DOF LPF controller incorporates a virtual spacing policy as its input. This control framework makes it easier for cars with different spacing strategies to create ad hoc platoons [19].

Vehicular Ad-Hoc Networks (VANETs) are used to simplify communication in order to handle the control challenge of leader-following vehicle platoons conforming to a constant-spacing policy. Platoon control is managed as a switched control system based on the connectivity state of each follower and the leader. A connection status matrix-dependent controller is a complex way to manage platooning vehicle dynamics, especially when communication reliability varies. The connection status matrix shows which platoon vehicles might reliably communicate at any given time [20].

2.2. Advanced control strategies

2.2.1. Consensus control

Neural networks, sliding mode method, and adaptive techniques are integrated in the consensus approach to control vehicle-following platoons when input saturation occurs. To estimate unknown nonlinear

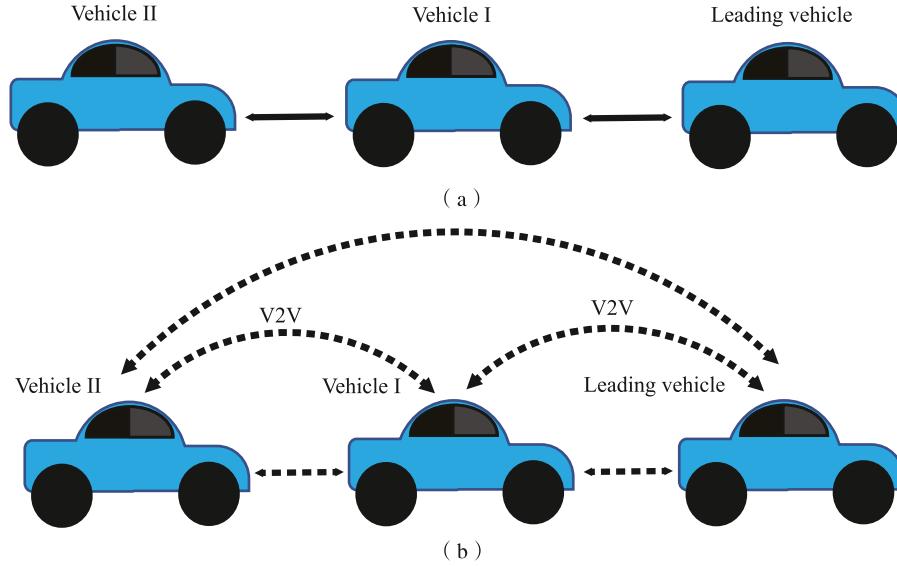


Fig. 2. The cruise control approach. (a) Adaptive Cruise Control (ACC). (b) Cooperative Adaptive Cruise Control (CACC).

functions obtained from the acceleration and velocity of the leader, Chebyshev neural networks are utilised. This approach addresses problems including input saturation, unmodeled dynamics, and disturbances to maintain the intended vehicle spacing and string stability. In automated platooning, multibody control complexity and unreliable wireless communication pose challenges. The consensus-based approach adapts to different scenarios and communication topologies, compensating for outdated information due to network losses and delays. It performs exceptionally well at maintaining platoon stability in the face of interference, delays, and fading circumstances, making drivers in the platoon feel more at ease and safer [21]. For vehicle platooning, a new control framework that guarantees vehicle convergence to specified velocities and intervehicle distances has been developed. The issue was presented as a high-order network consensus problem, and Lyapunov-Razumikhin functions were used to verify the problem's validity. This framework accommodates scenarios involving vehicles joining or leaving the platoon and experiencing communication failures. A unique control algorithm, rooted in distributed consensus, was developed to coordinate vehicles to maintain uniform inter-vehicle gaps. It considers the logical topology of communication and addresses issues such as delays and losses. Platooning problems in a network of dynamic systems affected by communication delays are handled as consensus problems. This technique is advanced and combines principles from control theory and network science [22]. Distributed control protocol synchronizes platoon vehicles. The Lyapunov-Razumikhin theorem ensures the platoon's stability and unanimity despite communication delays. The suggested distributed nonlinear consensus control technique is for connected vehicle (CV) platoons. It prioritises and accounts for car-following interactions while accommodating heterogeneous time delays and a reasonable longitudinal inter-vehicle distance. The control algorithm guarantees consensus among the CVs, preventing negative spacing errors and unreasonable acceleration/deceleration rates [23]. These approaches provide effective solutions for controlling vehicle platoons, ensuring stability, spacing control, and convergence despite input saturation, communication delays, and heterogeneous dynamics.

2.2.2. Smith predictor

In wireless network control systems, the Smith predictor technique is utilized for delay compensation to alleviate the negative impacts of time delays. The Smith predictor is a well-established control strategy that involves predicting future system behaviour based on past observations.

The goal of the delay compensation strategy using the Smith predictor in wireless network control systems is to reduce the effect of delays caused by the wireless channel. The classical Smith predictor and the adaptive Smith predictor are the two primary parts of the strategy. The classical Smith predictor utilizes a Markov model to estimate the network delay. By predicting the system's future behaviour based on this estimated delay, the controller can take proactive actions to compensate for the delay and maintain system stability. Conversely, channel delay information acquired by a shift register are included into the adaptive Smith predictor. This adaptive approach continuously updates the estimated delay, allowing the system to adapt to changing network conditions and improve the accuracy of delay compensation. Adding the Smith predictor greatly improves the performance of the wireless network control system when comparing the system's operation with and without the delay compensation method. This scheme effectively lessens the impact of delays, enhancing system stability and alleviating performance degradation, ultimately contributing to the overall effectiveness of the wireless network control system [24].

2.2.3. Model predictive control

CACC improves fuel efficiency, comfort, safety, and formation control in the context of intelligent cars. It makes use of a safety controller for emergencies and distributed model predictive control with multiple objectives (DMPCMO), a switching control approach intended for diverse platoons [25]. To improve economy and safety, the String Stable Model Predictive Cooperative Adaptive Cruise Control (MPCACC) system combines cooperative adaptive cruise control and model predictive control. MPCACC anticipates future states, optimizes control actions, enables communication and coordination among platoon vehicles, and achieves tighter spacing for improved traffic flow and reduced energy consumption [26]. Distributed model predictive control (DMPC) integrates CAVs into platoons. This complete algorithm includes DMPC platoon and space-making controllers. This integration simplifies merging, enables real-time control, and reduces computation needs [27]. DMPC achieves cooperative and flexible vehicle platooning in automated highway systems, solving local optimal control problems based on intervehicle communication. It improves performance, offers stability guarantees, efficient computation, and ensures collision-free requirements. Distributed model predictive controllers outperform centralized ones [28,29]. Model predictive control optimizes the performance of a connected vehicle platoon by adjusting speeds to minimize fuel consumption [30].

2.2.4. Optimal control

Optimal distributed control via sparse coprime factorizations enables efficient and decentralized control of vehicle platoons. It decomposes the platoon dynamics into smaller subproblems that each vehicle can independently address. This decentralized approach enhances scalability and fault tolerance and minimizes the need for extensive communication and coordination. Tasks such as regulating spacing, mitigating disturbances, and optimizing performance are assigned to each vehicle based on their coprime factorizations. This technique offers a promising solution for achieving efficient and robust platoon control, improving overall performance and stability [31].

2.2.5. H_∞ control

To address the acceleration disturbance of the lead vehicle in a platoon, the H_∞ controller is employed. To achieve string stability and ensure that there is no spacing error in the steady state, the controller is subject to certain requirements. The string stability, robustness, and tracking performance of a heterogeneous platoon with erratic vehicle dynamics are assessed using the H -infinity norm [32,33]. Platooning entails the synchronized motion of connected vehicles, and this study investigates its robustness by employing undirected topologies and a distributed H_∞ controller synthesis method. The primary focus of the analysis is to assess how well the platooning system performs and remains stable in various circumstances, such as changes in network topology and disruptions. The utilization of distributed H_∞ controller synthesis is implemented to improve coordination and safety in practical applications, including autonomous driving and intelligent transportation systems [34]. For a platoon of heterogeneous vehicles with unknown dynamics and communication delays, an H -infinity control scheme is developed to handle uncertainties and disturbances in actual driving scenarios. This method's main objective is to achieve string resilience, tracking performance, and stability. These goals are expressed in the linear fractional transformation format and are quantitatively evaluated using the H -infinity norm. The effectiveness of the control methodology is showcased through simulations, considering random parameters and external disturbances. This approach shows potential in improving road safety, highway efficiency, and traffic flow by forming platoons among highway vehicles operating under uncertain and dynamic driving conditions [35]. To effectively manage a heterogeneous vehicle platoon, a sophisticated control technique that considers variables such as aerodynamic drag, road gradients, and wireless communication delays is presented. The platoon formation process takes external disruptions, communication delays, and parameter uncertainty into account. A functional-based Lyapunov-Krasovskii H_∞ controller is developed for stability guarantee. This controller generates state-feedback control signals to stabilize vehicles and platoon strings. Unlike traditional string stability controllers, it tracks inter-vehicle spacing and maintains platoon string stability [36].

2.3. Control mode switching

2.3.1. Switched mode control

For platoon-level automation of autonomous vehicles, a novel robust acceleration tracking control system is developed to manage the longitudinal dynamics of vehicles. This system employs a multiple model switching structure to effectively handle uncertainties in vehicle dynamics. Multiple robust controllers are developed to address each model within the set by dividing the uncertainties into smaller segments. For higher-level platoon management, this method ensures more consistent and almost linear node dynamics. The system aims to enhance road traffic performance and safety, even in the presence of significant vehicle uncertainties [37].

2.3.2. Lateral control

Combining both lateral and longitudinal spacing schemes, the suggested control system exhibits excellent control performance and

flexibility to different curvature circumstances and speeds. Benefits of using a lateral control system for vehicle platoons include better traffic flow, fewer accidents, and increased fuel efficiency in addition to the benefits of traditional longitudinal control. In platooning, the two driving situations for which the lateral control techniques are intended are parallel movement and ordered movement. These driving conditions may also apply to other scenarios. The platoon's intended behaviour is achieved using certain lateral spacing control techniques, which are included into an integrated controller that includes an actuator control layer and a motion control layer [38].

2.3.3. L-platooning control

Vehicle platooning is an advanced automated driving technique that improves driving safety and fuel efficiency by enabling a group of cars to move closely together. Unfortunately, due to communication range limits, current platooning systems have trouble keeping up with extended platoons, especially when there are a lot of long-body heavy-duty vehicles involved. To address this issue, a new L-Platooning protocol has been developed, facilitating the seamless and rapid formation of extended platoons. The idea of a virtual leader, a tool that increases the original platoon leader's communication reach, is introduced by L-Platooning. The Virtual commander Quality Index (VLQI), which assesses a vehicle's performance as a platoon commander, is used as the basis for an election process to choose a virtual leader. Vehicles may effectively create extended platoons while precisely preserving the appropriate inter-vehicle spacing by using L-Platooning. Moreover, L-Platooning ensures the smooth integration and disengagement of vehicles within long platoons, adeptly managing join and leave maneuvers [39].

Each proposed system serves as a controller for the platoon and aims to address specific issues outlined in Table 1. Ensuring string stability within the platoon is the primary objective of these control strategies. However, in addition to string stability, there are several other critical issues that need to be addressed. These include topology changes, distance errors, headway time, spacing policies, time-varying delays, communication delays, and uncertainty in the communication channel. To maintain the platoon's overall performance and efficacy, these issues must be addressed in the controller designs. The key elements of the idea include establishing a reliable communication network, utilizing consensus-based control algorithms, implementing virtual coupling mechanisms, employing decentralized control, incorporating trajectory prediction algorithms, adapting control parameters, conducting robustness analysis, and exploring real-time optimization techniques.

3. Clustering

Clustering significantly improves Wireless Sensor Networks (WSNs) and VANETs' scalability, efficiency, and resource allocation. VANETs are composed of antenna-equipped vehicles that communicate while in motion, whereas WSNs are composed of resource-constrained, diminutive sensor nodes that gather environmental data. In VANETs, clustering divides vehicles into virtual groups or clusters based on their geographic proximity or other criteria. This helps in efficient communication, resource management, and addressing challenges related to limited resources and energy efficiency [40–43]. Clusters in VANETs can enhance routing scalability and reliability by forming hierarchical network structures based on spatial distribution and relative velocity. Clusters also support various applications in vehicular environments, such as accident detection, information dissemination, and entertainment. Similarly, in WSNs, clustering is employed to manage network resources and improve scalability and stability. In WSNs, sensor nodes with similar characteristics are grouped under a cluster head, efficiently managing the network by reducing the overhead associated with individual node communication. Clustering in WSNs enhances routing scalability, as fewer routing tables need to be maintained, and improves the network's ability to handle dynamic changes in topology. VANETs and WSNs face limited resources, energy efficiency, and scalability challenges.

Table 1
Comparison of performance measures in Platooning.

Serial number	Author	Controller used	Performance measures focused on	Performance measures not focused
1	Hossein Chehardoli et al. [74]	Adaptive control	Internal stability, string stability	A new control must be framed for each topology
2	Yuanheng Zhu et al. [12]	Adaptive optimal control, Cooperative Adaptive Cruise Control	String stability, improves traffic throughput	Need to stabilize distance errors
3	Mario di Bernardo et al. [21]	Consensus control	String stability of heterogeneous platoon	Minimize the time varying delays
4	Peng Liu et al. [37]	Distributed model predictive control	Cooperative control when joining and leaving the platoon	Uncertainty of communication channel must be considered
5	Gerrit J. L. et al. [41]	Decentralized controller, Cooperative Adaptive Cruise Control	String stability	Different communication structures must be compared
6	ZhongChang Liu et al. [40]	Decentralized controller	Cooperative control when joining and leaving the platoon	Communication delays need to be addressed
7	Yongfu Li et al. [15]	Distributed consensus control	Cooperative control by nonlinear method (considering time delays)	Time delays must be practically analyzed
8	Liwei Xu et al. [33]	H infinity controller	String stability, time delays in communication are unified	Comparison with other controllers not done
9	L. Xiao and F. Gao [34]	Sliding mode control	Platoon stability	Traffic flow, communication delays
10	C. Zhai et al. [23]	Smith predictor	String stability, accuracy of network delay compensation	Time delay compensation
11	S. Sabau et al. [38]	Optimal control	System stability	Overall traffic throughput
12	M. Won [30]	L-platooning control	Communication coverage	String stability

Clustering offers a solution to tackle these challenges in both types of networks. It aids in optimizing resource utilization, decreasing energy consumption, extending network lifetime, and improving overall network performance. Clustering is a technique commonly used in both VANETs and WSNs to enhance network performance, improve scalability, and address resource constraints. It provides comparable advantages in terms of efficient communication, scalable routing, and effective resource management in both network types [44,45].

3.1. Formation of clusters in VANET

Networks known as VANETs are formed by automobiles with antennas that communicate while moving, notably in highway platooning. VANETs let cars share road conditions, traffic updates, and safety warnings, improving road safety and efficiency. Clustering is a technique used to divide nodes into virtual groups, ensuring geographically adjacent nodes are placed in the same cluster. Evaluation of clustering focuses on metrics such as average message exchange, number of clusters, and cluster size. Platooning vehicles in highway systems offers benefits such as improved road capacity and energy efficiency. VANETs are used to enhance platoon performance, considering both VANET and platoon dynamics [46–48]. The disturbance-adaptive platoon (DA-Platoon) architecture considers disturbance scenarios and satisfies VANET constraints by adapting to changing conditions and determining the desired inter-platoon spacing. VANETs enable communication between vehicles and roadside infrastructure, contributing to road safety, smart transportation systems, and reduced accidents. They provide valuable data for optimizing traffic management and implementing adaptive traffic control systems [49]. VANETs also support smart city development by enabling efficient transportation management, energy efficiency, pollution reduction, and services like smart parking systems and autonomous driving. VANETs facilitate faster response times and optimized rescue operations through communication between emergency vehicles in emergency operations. Clustering is used in VANETs to improve network performance, addressing challenges related to limited resources and energy efficiency. However, non-optimized clustering can increase energy usage and reduce network scalability. Appropriate cluster sizes are necessary to maintain coverage and scalability. Routing protocols in extensive urban VANETs must exhibit reliability and scalability [50,51]. Clustering is suggested as a solution to enhance routing scalability and reliability by grouping vehicles according to spatial distribution and relative velocity. Clusters enhance routing and support various

applications such as accident detection, information dissemination, and entertainment. Accurate modeling of vehicle channels and clustering algorithm design choices are important considerations. This explores the role that information flow topology has in homogenous vehicular platoons' internal stability and scalability. Algebraic graph theory and stability criteria are utilized to understand the dynamics and behaviour of platoons operating in a rigid formation. Another framework considers ultra-cooperative communications in low signal-to-noise ratio channels, employing a scale-free topology and consensus principles to enhance intra cluster communication performance. In VANETs, clustering is essential for effective communication, enhanced network performance, and support for a range of applications in automotive settings [52].

VANETs serve as crucial infrastructure for smart transportation systems, accommodating the growing number of autonomous vehicles and supporting various activities that require internet connectivity in smart cities. VANETs facilitate communication among vehicles and roadside infrastructures, as depicted in Fig. 3, enhancing the intelligence of transportation systems. Moreover, VANETs enable on-board computers in predominantly stationary vehicles. By exchanging data regarding vehicle speed and location coordinates, VANETs aim to improve road safety regardless of whether infrastructure development is necessary.

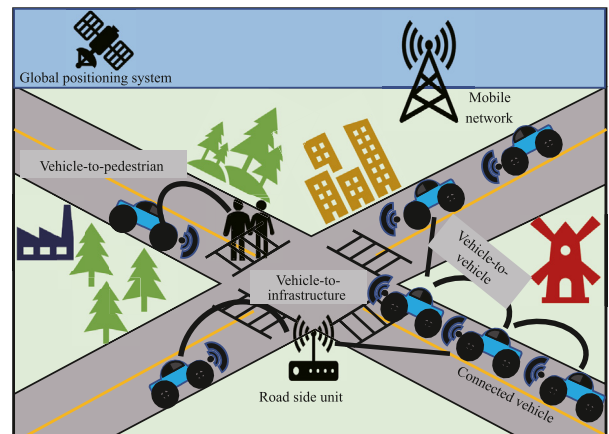


Fig. 3. Interaction and integration of smart transport systems.

3.2. Cluster formation in wireless sensor networks

In WSNs, grouping sensor nodes together enhances resource management, scalability, and durability. This strategy improves energy utilization, decreases communication overhead, and simplifies network data aggregation and processing. Clustering in WSNs aims to assign sensor nodes as cluster leaders, functioning as intermediates between them and the sink node or base station. In order to save network energy, these cluster heads gather and transmit data from member nodes to the sink node [53,54]. Clustering algorithms in WSNs typically consider factors such as node proximity, energy level, communication cost, and network topology to form clusters. A set of parameters, such as the highest remaining energy or a mix of energy and distance measures, are used to pick cluster heads during the process. The remaining sensor nodes join the cluster headed by the cluster leader who is closest to them. Following cluster creation, the cluster heads and the sink node are the main points of contact within the network. By aggregating sensor node data, cluster heads reduce energy consumption and network data transfer. To cut down on communication costs, cluster heads might examine local data first before transferring aggregated data to the sink node. Clustering in WSNs improves energy efficiency, scalability, and longevity. It reduces unnecessary data transmission and energy use by localizing data processing and aggregation. By separating troublesome nodes or clusters, clustering enhances network resilience and fault tolerance and simplifies network issue discovery and remediation [55]. Moreover, resource optimisation, energy conservation, and extending the network's operational lifetime all depend on clustering in WSNs. It is a key technique to achieving effective data collecting and processing in sensor networks with limited resources.

4. Control strategies in clustering

4.1. A link-aware, energy-efficient, and velocity-based Cluster-Tree approach for WSNs: the VELCT algorithm

Data collection is essential to WSN efficiency. For mobility, traffic, and end-to-end connection problems, current approaches are problematic. For Wireless Sensor Networks (WSNs), Velocity Energy-efficient and Link-aware Cluster-Tree (VELCT) takes end-to-end connection, traffic, mobility, latency, and coverage distance into consideration. The cluster head position determines VELCT's Data Collection Tree (DCT). The DCT data collecting node does not sense anything; it only accepts data packets from the cluster head and forwards them to the sink. This method effectively uses the DCT to decrease cluster head traffic, end-to-end latency, and energy consumption. The VELCT method creates a basic tree structure to reduce cluster head energy consumption and minimize frequent cluster creation while sustaining the cluster for a long time. Overall, the VELCT system is promising for increasing WSN data collecting [56].

4.2. The REAC-IN algorithm: an isolated node-based regional energy-aware clustering method for WSNs

In WSNs, clustering increases energy efficiency but necessitates choosing, assigning, and constructing cluster heads. REAC-IN combines regional energy awareness with isolated nodes to address these problems. The residual energy of each sensor and the regional average energy of all the sensors in each cluster are used to determine the weight of a cluster head (CH) in REAC-IN. REAC-IN aims to prevent nodes from getting isolated from CHs because connecting with the sink takes too much energy. Therefore, REAC-IN beats other clustering algorithms in network performance and energy economy [57].

4.3. Passive multihop communication in wireless networks

In VANETs, vehicle node grouping serves as an example of how effective cluster architecture is. However, traditional methods which

select cluster heads only on the basis of vehicle mobility might occasionally result in frequent switching and unstable connectivity amongst nodes. The multi-hop clustering methodology is employed by the Passive Multi-Hop (PMC) method to provide sufficient coverage and stability inside the cluster. During the cluster head selection phase, a priority-based neighbour-following approach is utilised to pick the best neighbour nodes for cluster membership, hence improving inter-cluster node stability and dependability. Stability-prioritization significantly improves clustering stability by identifying the most stable node in the N-hop range as the cluster head. To further enhance cluster resilience and reliability and address stability and reliability issues inside the cluster, a cluster merging technique is incorporated into the cluster maintenance phase [58].

4.4. Advancements in time synchronization, coordination control, and consensus tracking in wireless sensor networks and multi-agent systems

A time synchronisation technique with three steps—*intra-cluster time synchronisation*, *inter-cluster time synchronisation*, and *one-way intra-cluster time synchronization*—that uses a maximum consensus approach is used in a wireless sensor network. As a result, convergence is improved and communication traffic is reduced. The two components of the Clustered Consensus Time Synchronisation (CCTS) technique are *intra- and inter-cluster time synchronisation*. This approach incorporates clustering for quicker convergence and increased energy efficiency [59]. A distributed coordination control technique is presented to improve power generating reliability and lower costs in the context of linked DC microgrids. It provides a modified control formula to accomplish various objectives of cost and voltage control by using a hierarchical structure based on droop control. Fast convergence is guaranteed by a finite-time consensus algorithm [60]. For multi-agent systems with generic linear dynamics, a distributed predictive observer is utilised to estimate the consensus tracking error and offer an effective control input to address consensus tracking control. Convergence conditions are established using the generalised Nyquist criterion, proving delay-independence. The same criteria is used to compute the observer's gain matrix, which has a streamlined form that reduces it to a single parameter [61]. In summary, recent advancements in time synchronization, coordination control, and consensus tracking in wireless sensor networks and multi-agent systems have significantly improved network performance, energy efficiency, and coordination capabilities. These developments enhance these systems' overall operation and functionality for more robust and efficient applications in various domains.

4.5. Fuzzy based clustering algorithm

Networks may be used to simulate complex real-world situations, with nodes standing in for items and connections for the interactions between them. These networks have complex connection architectures and include important content data. The task at hand is efficiently locating significant clusters in these networks by using link and content data. FCAN, a fuzzy-based clustering technique, was created for complex networks to address this problem. To find fuzzy clusters that are highly relevant in terms of content and densely linked, the FCAN algorithm incorporates link and content information. It maximises each node's degree of participation in various clusters. In terms of performance, FCAN has potential scalability, precision, and computing efficiency [62].

4.6. Mobility-aware hierarchical clustering algorithms for mobile WSNs

WSNs are the foundation of the Internet of Things (IoT), but because of their limited resources, they require energy-efficient routing. By employing hierarchical or clustering-based routing, WSN scalability and energy efficiency have increased. However, issues including data loss, slower data rates, and higher energy consumption are brought about by the mobility of sensor nodes in Mobile WSNs (MWSNs). Using a three-

layer clustering hierarchy, mobility-aware hierarchical clustering algorithms have been created for WSNs in order to overcome these issues. Centralised gridding is used at both stages of the clustering hierarchy by the first method, known as the mobility-aware centralised clustering algorithm (MCCA). The second approach combines distributed clustering at the bottom layer with centralised gridding at the higher layer. It is called the mobility-aware hybrid clustering algorithm (MHCA). Improved data rates, reliable cluster formation, longer network lifetimes, and lower energy usage are all benefits of both methods for mobile sensor networks. Comparing the two approaches, the centralized clustering approach outperforms the hybrid clustering approach in various aspects of network performance. It is found to be superior in terms of network performance indicators. These mobility-aware clustering algorithms provide effective solutions for addressing the impact of node mobility in WSNs, ensuring efficient and reliable operation of the network [63].

4.7. Optimized cluster head selection scheme for vehicular communications

The Efficient Cluster Head Selection (ECHS) scheme is employed in vehicular communications to efficiently handle network resources. It involves clustering vehicles with similar characteristics, where a designated Cluster Head (CH) manages each cluster. However, the dynamic nature of vehicular networks makes choosing the right Cluster Heads (CHs) difficult. The ECHS scheme introduces specific conditions for cluster construction and CH selection to address these challenges. It prioritizes selecting a CH that centralizes the cluster, enabling longer connectivity with neighbouring vehicles. The scheme also ensures a proper distribution of clusters throughout the network by adjusting the distance between consecutive clusters. These conditions significantly enhance the effectiveness of clustering vehicles on the road. Through a comparative study, the ECHS scheme outperforms its equivalent on several measures. Overall, it is an effective method for cluster head selection in vehicular communications, resulting in improved network performance and resource management [64].

4.8. Mobile-to-cluster (M2C) approach for dynamic clustering in wireless sensor networks

New technologies that use mobile vehicles for wireless data gathering and charging extend the lifespan of WSNs. Using a dynamic clustering technique called mobile-to-cluster (M2C), the service process for sensor nodes and mobile vehicles is optimised. In contrast to conventional charging scheduling methods, M2C prevents sensor nodes from running out of energy before getting a charge by dynamically updating the network architecture based on local residual energy information. The M2C method actively selects sensor nodes with residual energy that is near to the predicted consumption as head nodes, therefore calculating the energy consumption associated with being a head node. This guarantees that head nodes with low remaining energy are given priority, which reduces the mobile vehicle's trip distance and increases charging efficiency. In comparison to existing approaches, M2C demonstrates significant improvements, reducing travel distance by approximately 10% and charging latency by about 50%. These outcomes underscore the effectiveness of the M2C approach in enhancing the performance of wireless sensor networks through optimized charging and efficient cluster head selection [65].

4.9. An optimised energy-efficient clustering algorithm for wireless sensor networks: the adaptive kernel density estimation (AKDE) algorithm

Effective energy conservation presents a significant challenge in Wireless Sensor Networks (WSNs), where the demand for high-quality services must be harmonized with the necessity to optimize energy usage for an extended network lifetime. Clustering routing algorithms have proven effective in facilitating data collection and transmission within WSNs. The random dispersion of nodes, however, may prevent conventional clustering algorithms—which aim for equal-sized clusters—from effectively

minimising energy use. An approach called Energy-Aware Adaptive Kernel Density Estimation (EAKDE) is presented in answer to this problem. To fairly distribute energy dissipation across cluster heads (CHs), the uneven clustering algorithm EAKDE was created. The priority of nodes vying to become CHs is determined by EAKDE using fuzzy logic. When it comes to network stability, longevity, and energy efficiency, EAKDE performs better than other well-known algorithms in a variety of settings [66].

4.10. Fuzzy Clustering protocol for balanced and energy-efficient cluster head selection in wireless sensor networks: LEACH-FC

WSNs seek to prolong the lifespan and expand the usefulness of their systems. The time it takes for the first node to run out of energy is one important element that determines how long a WSN can last. Utilizing clustering for cluster head selection proves to be an effective strategy in enhancing the lifespan of a WSN. However, existing clustering protocols based on probabilistic models may, at times, result in the selection of two cluster heads for nearby clusters, leading to decreased energy efficiency, particularly when the cluster head is situated at the cluster's periphery. To address this limitation, the LEACH-Fuzzy Clustering (LEACH-FC) protocol is employed. To extend the lifetime of the network, LEACH-FC uses fuzzy logic-based cluster head selection and cluster creation. Rather than using dispersed approaches, a centralised approach is used for both cluster head generation and selection. The vice cluster head, which likewise follows a centralised methodology, is determined by fuzzy logic. By guaranteeing a balanced energy load at every node, this eventually improves the WSN's dependability. LEACH-FC outperforms other algorithms in terms of energy usage and network lifespan improvement. Moreover, traditional clustering algorithms often result in unevenly distributed clusters, leading to unbalanced network loads. To address this issue, the LEACH-FC protocol introduces thresholds for cluster formation, ensuring the creation of balanced clusters. This further enhances the network's lifetime and reduces the node death rate compared to traditional approaches [67].

4.11. REPLACE: trust-based platoon service recommendation scheme for secure vehicular platooning

Fuzzy clustering algorithms have demonstrated significant promise in terms of accuracy, computing power, and scalability. The swift development of intelligent transportation networks has made room for novel strategies in highway design. The practice of driving closely together or vehicular platooning appears to be a viable way to reduce traffic, increase comfort, and boost safety. Nevertheless, the existence of misbehaving platoon head vehicles puts the safety of the whole platoon at risk, thus a system for user vehicles to recognise and steer clear of them is required. A trust-based platoon service recommendation system called REPLACE has been created to address this problem. The reputation system, which gathers and models user vehicle input on platoon head vehicle performance, is the central component of this method. An iterative filtering system is presented to reduce the effect of false information supplied by user automobiles to address problems linked to dishonest input. With security and resilience in mind, the REPLACE system is designed to survive a range of assaults in VANETs such as on-off attacks, ballot stuffing attacks, badmouth attacks, and newcomers. In the context of vehicular platooning, the approach offers user vehicles a safe and dependable way to avoid choosing misbehaving platoon head vehicles by implementing an iterative filtering mechanism and a reputation system [68].

4.12. Clustering-Tree Topology Control Algorithm (CTEF) for heterogeneous WSN energy management

The problem of energy consumption in heterogeneous wireless sensor networks (HWSNs) is addressed by the Clustering-Tree Topology Control Algorithm based on Energy Forecast (CTEF). By prioritising energy economy, load balancing, and connection quality, it maximises the

Table 2

Comparison of performance measures in the Clustering technique of wireless network.

Serial number	Author	Method used	Performance measures focused	Performance measures not focused
1	Farooq Aftab et al. [44]	Self-organization-based clustering	Stability of clusters, Reduces network congestion	Minimize the loss of energy
2	R Velmani et al. [54]	Velocity energy efficient and link aware cluster tree	Throughput, energy utilization, stable links	Delay in delivery of data packets
3	Degan Zhang et al. [29]	Passive Multi-hop clustering	Stability of clusters, reduces cost, reliability	Need to reduce inter-cluster interference
4	Huang Chang Ming et al. [75]	Prediction based clustering algorithm	Stable communication	Efficiency of network performance is to be increased
5	Mireille El Gheche et al. [36]	Multilayer network data clustering	Improves cluster performance	Modeling of node need to be considered
6	Masood Ahmad et al. [43]	Mobility based clustering, Minimum energy based clustering	Energy consumption, relative mobility	Maximum parameters need to be considered
7	Zhen Hong et al. [69]	Clustering tree topology algorithm	Packet loss rate, energy consumption	Network lifetime must be increased
8	Qinghai Yang et al. [77]	Multi-hop relay networks	Throughput, Coverage ratio	Minimize network congestion

network's longevity. Every round, CTEF estimates the average energy used by the network using an energy-efficient method. The central limit theorem and the normal distribution mechanism are used to compare the ideal and real residual energy to achieve this. Precise energy level prediction reduces the load on cluster heads, increasing the lifetime of the network. When it comes to packet reception at the base station and network lifetime, CTEF performs better than protocols like LEACH, EDFCM, and EDCS. It ensures better data transfer to the base station and a longer network lifespan. Through the use of energy prediction and cluster-based topology management, CTEF provides an efficient method for controlling energy in HWSN, enhancing overall performance and prolonging network longevity [69].

4.13. Hierarchical clustering-task scheduling policy for energy-efficient sensor networks

The goal of a hierarchical clustering technique for bipartite networks is to make complicated systems' higher-order organisations visible. The algorithm clusters bipartite networks using a multi-objective optimisation technique. The bipartite network is first converted to a monopartite network using a similarity index. Once the monopartite network has been converted, a multi-objective optimisation model is created to cluster it. The initial bipartite network is transformed into a hierarchical tree structure using an agglomerative technique. The effectiveness of the suggested approach for hierarchical clustering of bipartite networks is demonstrated by comparisons with a cutting-edge bipartite network clustering algorithm and a very effective hierarchical network clustering technique. Furthermore, an analysis of the algorithm-generated hierarchical trees reveals their biological significance, with possible consequences for the categorization of species [70,71]. For sensor node organisation in a clustered architecture, a hierarchical clustering-task scheduling strategy (HCSP) is presented to address the energy overhead related to global clustering operations in the conventional global round-based policy (GRBP). Although clustering has been shown to be useful for load balancing and increasing network longevity, GRBP's time-driven clustering technique results in energy overhead. HCSP offers a substitute by utilising a node-driven clustering methodology. Achieving a more adaptable, energy-efficient, and scalable clustering-task scheduling in comparison to GRBP is the main goal of HCSP. HCSP seeks to reduce energy consumption and increase network lifetime by reducing the clustering overhead, a major disadvantage of conventional clustering techniques [72,73].

4.14. Summary-exploring clustering strategies for enhanced platoon systems

In the realm of VANETs, platooning refers to the organized alignment of a cluster of vehicles that travel in proximity. This emerging concept leverages the capabilities of VANETs to enhance road capacity, optimize

energy efficiency, and streamline traffic flow. In platooning scenarios, vehicles maintain specific distances and speeds, often facilitated through advanced communication and control systems. Typically, vehicles in the platoon are equipped with wireless communication devices that enable real-time information exchange and coordination among them. VANETs play a pivotal role in supporting platooning operations by facilitating reliable and low-latency communication among the vehicles in the platoon. The vehicles exchange information about their position, speed, acceleration, and braking intentions, allowing coordinated platoon movements [74–76]. Platooning on VANETs offers several benefits. First, it improves traffic flow and reduces congestion by reducing the inter-vehicle gaps, effectively utilizing the available road space. Second, platooning enhances energy efficiency by minimizing aerodynamic drag and optimizing fuel consumption through synchronized movements. Third, it improves road safety by enabling faster reaction times and reducing the chances of accidents caused by human error. Many technologies and protocols enable platooning in VANETs. These include communication protocols like Dedicated Short-Range Communications (DSRC) or Cellular Vehicle-to-Everything (C-V2X) for smooth inter-vehicle communication, and Cooperative Adaptive Cruise Control (CACC), which enables platoon vehicles to synchronise their speeds with the lead vehicle [77–79]. Research in platooning on VANETs focuses on optimizing platoon formation, maintaining communication reliability, handling vehicle entry and exit from the platoon, and addressing challenges related to heterogeneous vehicle characteristics and road conditions. Overall, platooning on VANETs holds great promise for improving road capacity, energy efficiency, and safety. This field is currently undergoing significant research and development, holding the promise of revolutionizing highway travel and fostering more efficient and sustainable transportation systems.

The clustering techniques from 4.1 to 4.13 represents the implementation in network nodes which functions similarly as that of platooning. Therefore, the goal is to use clustering approaches to vehicle platooning in order to ensure excellent communication within a platoon platform and stability regardless of the type (heterogeneous vehicles) and number of vehicles. In Table 2, the performance parameters such as energy, throughput, latency, stable connection focused in the many of proposed systems. In addition to that, control strategy for a stable connection under high mobility nodes is to be addressed. Furthermore, the constraints such as road segment topology, clustering on different control strategies, and disconnections are to be focused.

5. Conclusion

An overview of wireless network applications in the context of vehicle platooning is given in this article. It explores various controllers and their corresponding control algorithms used for platooning formation,

comparing their performance. The study also discusses different methods to achieve efficient platoon performance. Additionally, the paper identifies and examines clustering techniques that are like platooning concepts. It emphasizes the potential of VANETs to enable networking among vehicles, bridging control strategies, and networking concepts for dynamic vehicle networks. Moreover, the paper highlights the consideration of road topology and mobile node constraints in clustering to form platoons. Future research directions include investigating the impact of network parameters on different control strategies of platoons and vice versa, establishing a networked vehicle platoon.

CRedit authorship contribution statement

M. Nandhini: Writing – review & editing, Writing – original draft, Validation. **M. Mohamed Rabik:** Writing – review & editing, Supervision.

Declaration of competing interest

There is no conflict of interests for this article for publication.

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